Yakkala Amarnath Karthik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.93
Bhashyam Junior College	Class XII (Board of Intermediate Education AP)	2023	98.10%
NARAYANA (EM) SCHOOL	Class X (AP SSC)	2021	99.00%

WORK EXPERIENCE

Expedia Group | Data Engineer Intern

- Developed **COMET**, a Slack-native AI operations assistant using Python and **ReAct-based reasoning**, integrating 20+ operational tools for diagnosing and resolving Airflow pipeline incidents.
- Integrated and debugged **MCP-backed tools and agent workflows** across Airflow, EGDQ, Querybook/Trino, and Datadog, enabling multi-step, tool-grounded incident diagnosis directly from Slack.
- Implemented **fuzzy DAG-name recognition and safe action handling**, enabling typo-tolerant pipeline identification while enforcing human confirmation for production actions such as retry, trigger, clear, pause, and unpause.

PROJECTS

RepoLens - AI-Powered Repository Analysis Agent

- Built an AI-powered repository analysis agent that **investigates GitHub and local codebases**, answers natural-language questions, and generates **line-level source citations**.
- Implemented an **evidence-first retrieval pipeline** combining repository scanning, AST parsing, semantic chunking, hybrid vector-lexical search, and LLM reasoning to generate grounded codebase-level answers.
- Designed a modular Python architecture with **Qdrant** vector indexing, repository fingerprinting, deterministic code-analysis tools, and **configurable LLM backends**.

FluxGate - Fault-Tolerant Reverse Proxy & Load Balancer

- Built a concurrent **reverse proxy and load balancer** supporting **HTTP (L7)** and **TCP (L4)** traffic with **Weighted Round Robin** and **Least Connections** routing.
- Implemented **health checks**, **request retries**, **circuit breaking**, **rate limiting**, and **configurable timeouts** to maintain availability under backend failures and overloaded conditions.
- Added **Prometheus-based observability** and benchmarked routing behavior, failure recovery, and request latency across multiple backend servers.

Nearest Neighbor Search using Generalized Hyperplane Tree

- Developed a C++ tool to find nearest neighbors in **large n-dimensional datasets**, efficiently handling very large numbers of points **using a GH-Tree**.
- Optimized algorithmic design with triangle inequality pruning to **skip up to 60% of distance checks**, boosting performance on large datasets.
- Implemented **recursive hyperplane partitioning** with dual-pivot selection and configurable leaf sizes for efficient nearest-neighbor search.

SKILLS

- **Programming languages:** Python, C++, SQL, C

- **Core CS**: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, DBMS

- **AI/ML:** Machine Learning, LLMs, AI Agents, RAG, ReAct, LangChain, LangGraph, Embeddings, Semantic Search
- **ML Libraries/Frameworks:** TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, OpenCV
- **Databases:** MySQL, Qdrant
- **Tools:** Git/GitHub, Confluence, MATLAB, LaTeX

RELEVANT COURSES

- **Core Computer Science Courses:** Introduction to Programming, Data Structures and Applications, Database Management Systems 1
- **AI/ML:** Introduction to AI/ML, Pattern Recognition and Machine Learning, Statistical Inference, Predictive Modelling
- **Mathematics & Optimization:** Probability and Random Processes, Matrix Theory, Linear Optimization, Convex Optimisation

POSITIONS OF RESPONSIBILITY

- **Informals Coordinator, Elan & nVision**

Organized Nexus with 75+ teams, introduced Mystery Rooms and sponsored wooden games, and guided ~25 deputy coordinators as Informals Coordinator.

- **Media and PR Head, National Service Scheme**

Led Media and PR for NSS, managing outreach campaigns, social media, media coverage, and merchandise designs, boosting engagement and strengthening the organization's brand.

- **Sunshine Mentor, IIT Hyderabad**

Mentored a group of junior students by providing academic guidance, personal support, and organizing sessions to help them adapt smoothly to campus life.

EXTRACURRICULAR

- **Represented IIT Hyderabad** as a dancer at **Inter IIT Cultural in 2024 and 2025**.
- Skilled in **video editing, photography, visual content creation** for events, promotions, and social media.